

DS Practical 3

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Stack Application:

Write a program to:

a. Implement a stack using an array.

```
import array as arr
stack1 = arr.array('i')

print(stack1)

def push():
    print("Push Operation")
    ele = int(input("Enter your element: "))
    stack1.append(ele)
    print(stack1)

def pop():
    print("Pop Operation")
    if(isEmpty()==0):
        print("Stack Underflow")
    else:
        stack1.pop()
        print(stack1)

def peek():
    print("Peek Operation")
    if(isEmpty()==0):
        print("Stack Empty")
    else:
        print(stack1[-1])

def isEmpty():
    if len(stack1)==0:
        return 0

num = 0

while(num<5):
    print("1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit ")
    num = int(input("select ch:"))
```

```

if(num==1):
    push()
elif(num==2):
    pop()
elif(num==3):
    peek()
elif(num==4):
    isEmpty()
else:
    print("program closed")
    print("S009")

```

```

===== RESTART: C:/Users/mvluc/Downloads/F010/P3.py
array('i')
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:0
program closed
S009
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit |
select ch:1
Push Operation
Enter your element: 11
array('i', [11])
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:1
Push Operation
Enter your element: 12
array('i', [11, 12])
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:1
Push Operation
Enter your element: 13
array('i', [11, 12, 13])
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:2
Pop Operation
array('i', [11, 12])
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:3
Peek Operation
12
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:0
program closed
S009

```

```

import array as arr

stack1 = arr.array('i')
si = int(input("enter the size of stack:"))
print(stack1)

def push():
    print("Push Operation")
    if len(stack1) == si:
        print("Stack overflow")
    else:
        ele = int(input("Enter your element: "))
        stack1.append(ele)
        print(stack1)

def pop():
    print("Pop Operation")
    if isEmpty() == 1:
        print("Stack Underflow")
    else:
        stack1.pop()
        print(stack1)

def peek():
    print("Peek Operation")
    if isEmpty() == 1:
        print("Stack Empty")
    else:
        print(stack1[-1])

def isEmpty():
    if len(stack1) == 0:
        return 1
    return 0

while True:
    print("1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit ")
    num = int(input("sselect ch:"))

    if num == 1:
        push()
    elif num == 2:
        pop()
    elif num == 3:
        peek()
    elif num == 4:
        if isEmpty() == 1:

```

```

        print("Stack is empty")
    else:
        print("Stack is not empty")
    elif num == 5:
        print("program closed")
        print("S009")
        break
    else:
        print("Invalid choice")

enter the size of stack:3
array('i')
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:1
Push Operation
Enter your element: 11
array('i', [11])
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:1
Push Operation
Enter your element: 12
array('i', [11, 12])
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:1
Push Operation
Enter your element: 13
array('i', [11, 12, 13])
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:2
Pop Operation
array('i', [11, 12])
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:3
Peek Operation
12
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:4
Stack is not empty
1.Push 2.Pop 3.Peek 4.isEmpty 5.Exit
select ch:5
program closed
S009

```

b. Convert an infix expression to postfix notation using a stack.

